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A DEVICE FOR GASIFICATION OF FEEDSTOCK

Abstract

A gasification device (**100**) comprises an exothermic chamber (**130**) provided with a combustion zone (**131**) that is configured to perform combustion to produce process heat, a first endothermic chamber (**110**) adapted to perform gasification of feedstock; and a second endothermic chamber (**120**) adapted to subject feedstock to an endothermic break-down reaction. The gasification device (**100**) is configured such that the first endothermic chamber (**110**) uses combustion process heat to perform the gasification process. The gasification device (**100**) is configured such that the second endothermic chamber (**120**) uses combustion process heat to subject feedstock to the endothermic break-down reaction. The gasification device further comprises at least one heat-transfer rod (**140**) adapted to transfer combustion process heat from the combustion zone (**131**) of the exothermic chamber (**130**) to the first endothermic chamber (**110**) and to the second endothermic chamber (**120**).

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Background/Summary

TECHNICAL FIELD
[0001] The present disclosure relates to a device for gasification of feedstock.
BACKGROUND OF THE INVENTION
[0002] Gasification devices are used for gasifying feedstock. The feedstock may be biodegradable material that comprises a composition of carbon or carbohydrates containing substance, Hydrogen, Nitrogen, Calcium etc. The biodegradable material or biomass composition is generally found in organic materials like wood, plastic, pesticides, herbicides, pathogens, paints, contaminated solvents, residues from the paper and cellulose production, coal, tar, tar sand to name a few. This kind of feedstock is collected from homes, hospitals, power plants, oil refineries etc. The gasification devices make use of the feedstock to produce fuel in an environmentally friendly manner. The gasification devices produce fuel like a synthesis gas or a producer gas. Synthesis gas (Syngas) comprises carbon monoxide (CO) and hydrogen (H2), which is the product of steam or oxygen gasification. Producer gas is the mixture of gases produced by the gasification of organic material such as biomass. Producer gas is composed of carbon monoxide (CO), hydrogen (H2), oxygen (O2), carbon dioxide (CO2) and typically a range of hydrocarbons such as methane (CH4) with nitrogen from the air. This fuel has applications including, but not limited to, operating gas, steam, hydrogen engines.
[0003] EP1187892 discloses a facility for producing combustible gas from carbon-containing, in particular biogenic feed materials by allothermic steam gasification. The facility having the following features a pressure-supercharged fluidized-bed gasification chamber with a pressure-tight lock for supplying the feed materials, which are to be gasified. The facility is further having a filter chamber, which is connected to the fluidized-bed gasification chamber via a connecting channel. The facility is further having an external heat source. The facility is further having a heat-pipe arrangement, which takes up heat from the external heat source and gives it off to the gasification bed in the fluidized-bed gasification chamber.
Overview
[0004] Thus, there is a need to overcome drawbacks of prior art. In particular, there is a need to overcome drawbacks of conventional gasification devices. The various aspects of the present invention overcome drawbacks of the prior art. The following presents a simplified overview in order to provide a basic understanding of one or more aspects of the invention. This overview is not an extensive overview of the invention, and is neither intended to identify key or critical elements of the invention, nor to delineate the scope thereof. Rather, the primary purpose of the overview is to present some embodiments of the invention.
[0005] According to the invention in an aspect a device for gasification of feedstock is provided. The device comprises an exothermic chamber, a first endothermic chamber and a second endothermic chamber. The exothermic chamber is provided with a combustion zone that is configured to produce combustion process heat. The first endothermic chamber is adapted to perform a gasification process, in particular, an allothermic gasification process. The second endothermic chamber is adapted to have the feedstock perform endothermic break-down reactions. The device is configured such the second endothermic chamber and the first endothermic chamber use the process heat. The device further comprises at least one heat-transfer rod adapted to transfer combustion process heat from the combustion zone of the exothermic chamber to the first

endothermic chamber and to the second endothermic chamber. At least one effect can be that the rods are better than hollow tubes and more particularly, the rod are easy to maintain, high durability and better heat transfer properties.
[0006] In some embodiments, the at least one heat-transfer rod extends into the first endothermic chamber and into the second endothermic chamber. In some embodiments, the at least one heat-transfer rod is arranged inside the exothermic chamber. In some embodiments, the at least one heat-transfer rod, arranged inside the exothermic chamber, extends into the first endothermic chamber and into the second endothermic chamber.
[0007] In some embodiments, the at least one heat-transfer rod comprises a solid heat pipe, a hollow heat pipe or a convection heat pipe.
[0008] In some embodiments, at least two of the second endothermic chamber, the first endothermic chamber and the exothermic chamber are arranged above one another.
[0009] In some embodiments, at least one of the second endothermic chamber and the first endothermic chamber is arranged inside the exothermic chamber.
[0010] In some embodiments, the first endothermic chamber is a reformer.
[0011] In some embodiments, the reformer comprises a fluidized reformer-bedchamber.
[0012] In some embodiments, the first endothermic chamber includes a reformer inlet configured to be coupled to a source of steam.
[0013] In some embodiments, the device further comprising a steam generator configured to receive water and to generate steam from the water. In some embodiments, the steam generator comprises a steam outlet connected to the reformer inlet. In some embodiments, the device further comprising a steam generator configured to receive water and to generate steam from the water, wherein the steam generator comprises a steam outlet connected to the reformer inlet.
[0014] In some embodiments, the at least one heat-transfer rod is further adapted to transfer the combustion process heat from the exothermic chamber to the steam generator.
[0015] In some embodiments, the second endothermic chamber is a pyrolysis reactor.
[0016] In some embodiments, the pyrolysis reactor comprises a fluidized reactor-bedchamber.
[0017] In some embodiments, the combustion zone of the exothermic chamber is configured to combust fuel to produce the process heat.
[0018] In some embodiments, the exothermic chamber is configured to combust feedstock. In some embodiments, the exothermic chamber is comprises an inlet configured to receive feedstock.
[0019] In some embodiments, the exothermic chamber further comprises a bottom inlet configured to be coupled to an oxygen source. In some embodiments, the exothermic chamber further comprises a bottom outlet configured to release ash from the exothermic chamber. In some embodiments, the exothermic chamber further comprises a bottom inlet configured to be coupled to an oxygen source and a bottom outlet configured to release ash from the exothermic chamber.
[0020] The independent claim defines the invention in one aspect. The dependent claims state selected elements of embodiments according to the invention. It is to be noted that elements of these embodiments may be combined with each other unless specifically noted to the contrary.
[0021] This overview is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. This overview is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter. Those skilled in the art will recognise additional features and advantages upon reading the following detailed description, and upon viewing the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] FIG. **1** is a drawing that schematically illustrates a sectional view of a gasification device according to some embodiments.

[0023] FIG. **2** is a drawing that schematically illustrates a sectional view of a gasification device according to some embodiments.

DETAILED DESCRIPTION OF THE INVENTION

[0024] Below, embodiments, implementations and associated effects are disclosed with reference to the drawings that illustrate views of some embodiments. It should be noted that views of exemplary embodiments are merely to illustrate selected features of the some embodiments. In particular, cross-sectional views are not drawn to scale and dimensional relationships of the illustrated structures can differ from those of the illustrations. As used herein, like terms refer to like elements throughout the description.

[0025] It is to be understood that the features of various embodiments described herein may be combined with each other, unless specifically noted otherwise. In some instances, well-known features are omitted or simplified to clarify the description of the exemplary implementations. The order in which the embodiments/implementations and methods/processes are described is not intended to be construed as a limitation, and any number of the described implementations and processes may be combined.

[0026] FIG. **1** is a drawing that schematically illustrates a sectional view of a gasification device **100** for gasification of feedstock according to an aspect of the invention. In some embodiments, the gasification device **100** comprises an exothermic chamber **130**, a first endothermic chamber **110** and a second endothermic chamber **120**. In some embodiments (not shown), the first endothermic chamber comprises the second endothermic chamber. In some embodiments (not shown), the device comprises an exothermic chamber and a main endothermic chamber that comprises a first endothermic chamber and a second endothermic chamber. At least one effect can be that steps of a gasification process can take place within the gasification device **100**, i.e., performing endothermic break-down reactions on feedstock to produce a gas and performing exothermic reactions to produce process heat, wherein the process heat can be used for carrying out the endothermic reactions.

[0027] In some embodiments, the gasification device **100** has at least two of the second endothermic chamber **120**, the first endothermic chamber **110** and the exothermic chamber **130** arranged above one another. At least one effect can be that heat transfer by convection is improved in comparison with alternate arrangements. Further, gravity can be used in providing feedstock to the first endothermic chamber **110**, to the second endothermic chamber **120** and/or to the exothermic chamber **130**. Thus, the efficiency of the gasification device **100** can be improved. Further effect can be that the gasification device **100** is kept structurally simple. In particular, the gasification device **100** can efficiently allow the material to flow, reducing loss of heat or resources to the surrounding, from the second endothermic chamber **120** to the first endothermic chamber **110**, from the second endothermic chamber **120** to the exothermic chamber **130**, and/or from the first endothermic chamber **110** to the exothermic chamber **130**.

[0028] In some embodiments, the gasification device **100** has at least one of the first endothermic chamber **110** and the second endothermic chamber **120** arranged inside the exothermic chamber **130**. In some embodiments, the gasification device **100** has both the first endothermic chamber **110** and the second endothermic chamber **120** arranged inside the exothermic chamber **130**. In some embodiments, the exothermic chamber **130** is configured to have the process heat produced inside the exothermic chamber **130** by combustion of feedstock. At least one effect can be that the first endothermic chamber **110** and the second endothermic chamber **120** carry out the respective endothermic reactions using process heat produced by the exothermic chamber **130**. Thus, the need for an external heat source is reduced. Accordingly, the first endothermic chamber **110** can be configured to perform endothermic reactions without use of any additional exothermic chamber. Likewise, the second endothermic chamber **120** can be configured to perform endothermic reactions without use of any additional exothermic chamber.

[0029] The gasification device **100** for gasification of feedstock further comprises a plurality of heat-transfer rods **140**. In one embodiment (not shown), the gasification device comprises a single heat-transfer rod. In some embodiments, the plurality of heat-transfer rods **140** extend from the combustion zone of the exothermic chamber **130** to the first endothermic chamber **110**. Thus, the plurality of heat-transfer rods **140** can transport process heat generated in the combustion zone **131** of the exothermic chamber **130** to the first endothermic chamber **110**. In some embodiments, the plurality of heat-transfer rods **140** extend from the combustion zone of the exothermic chamber **130** to the second endothermic chamber **120**. Thus, the plurality of heat-transfer rods **140** can transport process heat generated in the combustion zone **131** of the exothermic chamber **130** to the second endothermic chamber **120**. In some embodiments, the plurality of heat-transfer rods **140** extends from the combustion zone of the exothermic chamber **130** through the first endothermic chamber **110** to the second endothermic chamber **120**. Thus, the plurality of heat-transfer rods **140** can transport process heat generated in the combustion zone **131** of the exothermic chamber **130** to the second endothermic chamber **120** and to the first endothermic chamber **110**.

[0030] In the example illustrated in FIG. **1**, the gasification device **100** is configured such that both, the second endothermic chamber **120** and the first endothermic chamber **110**, use the process heat produced in the exothermiczone **131** of the exothermic chamber **130**. The process heat, generated in the combustion zone **131** of the exothermic chamber **130** and produced at a temperature of at least 1100° C., for example, at a temperature in a range of from 1100° C. to 1600° C., in particular at a temperature in a range of from 1200° C. to 1400° C., rises up to the first endothermic chamber **110** and then to the second endothermic chamber **120**. In some embodiments, the gasification device **100** is configured to additionally use convection, wherein, for example, hot gas rises from the combustion zone **131** of the exothermic chamber **130** along an outside surface of a wall of the first endothermic chamber **110** to exchange heat with the wall of the first endothermic chamber **110** and transfer energy to the first endothermic chamber **110**. Likewise, the hot gas can rise further from the first endothermic chamber **110** along an outside surface of a wall of the second endothermic chamber **120**. Thus, a first portion of the process heat is transported to the first endothermic chamber **110** operating at the temperature of at least 700° C. As the process heat rises up in the gasification device **100**, heat transport takes place to the outside of the device and the temperature of the process heat decreases. Further, a second portion of the process heat is transported to the second endothermic chamber **120** operating at the temperature of at least 350° C. Thus, process heat produced in the combustion zone **131** of the exothermic chamber **130** heats the outside of both, the first endothermic chamber **110** and the second endothermic chamber **120**. Some of the process heat reaching the second endothermic chamber **120** enters into the performance of pyrolysis.

[0031] At least one effect of the above configuration can be that the device can be configured so as to achieve heat transfer as required. The gasification device **100** achieves an improved heat transfer from the combustion zone **131** of the exothermic chamber **130** to the first endothermic chamber **110** and/or to the second endothermic chamber **120**, since it uses at least two paths of heat transfer: A first path is provided by convective heat transfer that takes place from the combustion zone **131** of the exothermic chamber **130** to the first endothermic chamber **110** and/or to the second endothermic chamber **120**. A second path is provided by conductive heat transfer from the combustion zone **131** of the exothermic chamber **130** to the first endothermic chamber **110** and/or to the second endothermic chamber **120** that takes place heat transfer in the at least one heat-transfer rod **140** that extends from the combustion zone **131** of the exothermic chamber **130** through the inside of the first endothermic chamber **110** to the inside of the second endothermic chamber **120**.

[0032] Having regard to the exothermic chamber **130** of the gasification device **100**, the combustion zone **131** is adapted to produce process heat in the combustion zone **131**. In the combustion zone, the exothermic chamber **130** is configured to withstand a temperature that can occur during combustion. For example, the exothermic chamber **130** is configured to withstand a temperature of 800° C., of 900° C., or of 1000° C. In some embodiments, the exothermic chamber **130** is configured to withstand a temperature of 1100° C.

[0033] In some embodiments, the exothermic chamber **130** is configured to receive the feedstock for combustion in the form of pyrolysis vapors, gases and/or pyrolysis coke or ash from the first endothermic chamber **110** and/or from the second endothermic chamber **120** and/or directly from an external source **170** to produce process heat.

[0034] The exothermic chamber **130** comprises a first inlet **172** adapted to receive feedstock for combustion in the combustion zone **131** from an external source of feedstock. Further, in one example, a second inlet **174** is provided to the exothermic chamber **130** in proximity to the combustion zone **131**. The second inlet **174** is adapted to receive feedstock from the first endothermic chamber **110** and/or from the second endothermic chamber **120**. In some embodiments (not shown), the exothermic chamber **130** has one inlet to commonly receive feedstock for combustion from any one or any combination of an external source, the first endothermic chamber **110** and/or the second endothermic chamber **120**. Thus, feedstock can continuously be supplied to the exothermic chamber **130** for combustion in the combustion zone **131**. The exothermic chamber **130** can be configured to receive the feedstock for combustion as intermediate products in any form, like a mixture of pyrolysis vapors, gases and/or pyrolysis coke or ash, from the first endothermic chamber **110** and/or from the second endothermic chamber **120** to produce process heat. Thus, feedstock can continuously be supplied to the combustion zone **131** for combustion in the exothermic chamber **130** to produce process heat.

[0035] In some embodiments, the exothermic chamber **130** comprises a bottom inlet **132**. In some embodiments, the bottom inlet **132** is configured to couple the combustion zone **131** of the exothermic chamber **130** to a source of combustion air. Thus, oxygen in the combustion air can burn fuel in the combustion zone **131** of the exothermic chamber **130**. In some embodiments, the exothermic chamber **130** is configured to develop a fluidized bed in the combustion zone **131**. Thus, the bed of the exothermic chamber **130** in the combustion zone **131** can be fluidized. During combustion, feedstock can float on the fluidized bed in the combustion zone **131** of the exothermic chamber **130**. Thus, the feedstock can be evenly distributed around the at least one heat-transfer rod **140**. In an embodiment of the gasification device that comprises a plurality of heat-transfer rods **140**, the feedstock can be evenly distributed between the plurality of heat-transfer rods **140**.

[0036] In some embodiments, the exothermic chamber **130** comprises a bottom outlet **134**. In some embodiments, the bottom outlet **134** is configured to release ash from the exothermic chamber **130**. At least one effect can be that the residue of feedstock after combustion, gasification and pyrolysis is removed from the gasification device **100**.

[0037] In some embodiments, the exothermic chamber **130** is configured to combust a variety of fuels. The combustion zone **131** of the exothermic chamber **130** may be configured as a multi-fuel combustion zone capable of receiving and combusting the variety of fuels and/or configured with a multi-fuel burner, so that the variety of fuels can be selectively provided to the combustion zone **131** of the exothermic chamber **130**. Feedstock can be anything that comprises carbon and burns. For example, feedstock can comprise one or more from a group of fermentable, biomass-containing residual materials consisting of sewage sludge, biowaste or food waste, farm manure (liquid manure, dung), previously unused plants as well as plant parts (for example catch crops, plant residues and the like), specifically cultivated energy crops (renewable raw materials). Preferably, an amount of corrosive components, such as chloride or sulfide, in the feedstock is kept low or even to a minimum. One effect of keeping the amount of corrosive components in the feedstock low is that the corrosion of the chamber is reduced.

[0038] In particular, the exothermic chamber **130** can be configured to receive feedstock for combustion in the combustion zone **131** of the exothermic chamber **130**. For example, the feedstock includes solid phase feedstock and/or gaseous/vaporized phase feedstock. Thus, the exothermic chamber **130** can be used to perform combustion on any phase of feedstock, thereby reducing, for example, a need for additional fuel for combustion. Further, a need for filters can be reduced that are conventionally used to separate feedstock suitable for combustion from other material. In some embodiments, the combustion zone **131** of the exothermic chamber **130** is adapted to receive process products and/or by-products of gasification obtained from pyrolysis performed in the second endothermic chamber **110**. In some embodiments, the combustion zone **131** of the exothermic chamber **130** is adapted to receive products and/or by-products of gasification performed in the first endothermic chamber **110**. Thus, use of feedstock in the gasification device **100** can be optimized.

[0039] In some embodiments, the combustion zone **131** of the exothermic chamber **130** can be configured to combust additional feedstock obtained directly from another source, like intermediate products of a pyrolysis reaction obtained from the external source **170** to produce process heat. Thus, a need to provide feedstock for combustion from the first endothermic chamber **110** and the second endothermic chamber **120** to the combustion zone **131** of the exothermic chamber **130** is reduced. Thus, the efficiency of the gasification device **100** to produce process heat is improved.

[0040] In some embodiments, the exothermic chamber **130** can be adapted to allow a variation of a configuration according to operational parameters of the gasification device **100**, like the input of feedstock from any one or any combination of the inlet **172** and the inlet **174**, temperature, pressure etc.

[0041] In some embodiments, the first endothermic chamber **110** of the gasification device **100** is adapted to perform an allothermic gasification process. In particular, the first endothermic chamber **110** can be configured as a reformer. For example, the first endothermic chamber **110** can comprise a multi-material allothermic gasification-reforming reactor. At least one effect can be that the gasification can be performed inside the gasification device **100** directly upon pyrolysis. In some embodiments, the first endothermic chamber **110** is configured to develop a fluidized bed. An effect of making the bed of the first endothermic chamber **110** fluidized can be, as the feedstock floats on the bed of the first endothermic chamber **110**, that the feedstock is evenly distributed between the rods **140**.

[0042] In some embodiments, the first endothermic chamber **110** comprises a reformer inlet **116**. The reformer inlet **116** is configured to receive steam. The reformer inlet **116** couples the first endothermic chamber **110** to a source of steam. At least one effect of steam in the first endothermic chamber **110** can be that the steam is used to produce gas preferably, but not limited to, synthesis gas (CO+H2), i.e., $CH4+H2O \rightleftarrows CO+3\ H2$. In some embodiments, the first endothermic chamber **110** comprises a first outlet **114**. In some embodiments, the first endothermic chamber **110** further comprises a second outlet **115**. An effect can be to transfer the feedstock from the first endothermic chamber **110** to the exothermic chamber **130**.

[0043] In some embodiments, the first endothermic chamber **110** comprises a gas outlet **112**. The gas outlet **112** is configured to supply the produced gas after, preferably synthesis gas combination of carbon monoxide and hydrogen, outside the gasification device **100**. In some embodiments, the gas outlet **112** is coupled with a gas container (not shown) to supply the produced gas, preferably synthesis gas combination of carbon monoxide and hydrogen, outside the gasification device **100** directly to the container. At least one effect can be that the gas outlet **112** couples the first endothermic chamber **110** to the gas container (not shown) for further applications, like running engines.

[0044] In some embodiments, the first endothermic chamber **110** performs gasification process on feedstock to produce a gas. In some embodiments, the first endothermic chamber **110** is configured to perform gasification process on feedstock to produce a synthesis gas CO+H2 or a producer gas

CO+H2 or CO2+H2. In some embodiments, the first endothermic chamber **110** performs gasification process on feedstock to produce a hydrogen rich synthesis gas. The first endothermic chamber **110** operates at a temperature of greater than 700° C. or in the range of from 700° C. to 1100° C. Thus, gasification of the biodegradable material can takes place inside the first endothermic chamber **110** by reacting the feedstock material at temperatures in an endothermic reaction.

[0045] In some embodiments, the first endothermic chamber **110** may be modified according to the operational parameters of the gasification device **100**, like temperature, pressure etc.

[0046] The second endothermic chamber **120** is adapted to have the feedstock perform endothermic break-down reactions. In particular, the second endothermic chamber **120** is configured as a pyrolysis reactor. In some embodiments, the second endothermic chamber **120** comprises an inlet **122**. The inlet **122** is configured to receive feedstock from a storage outside the gasification device **100**. In some embodiments, the second endothermic chamber **120** comprises an upper outlet **124**. In some embodiments, the second endothermic chamber **120** comprises a lower outlet **125**. Thus, feedstock can be transferred from the second endothermic chamber **120** to the exothermic chamber **130**. In some embodiments, the second endothermic chamber **120** is configured to develop a fluidized bed. Thus, solid particles of feedstock can flow in the fluidized bed, i.e., the solid and fluid parts of feedstock can be transported in the fluid within the gasification device **100**.

[0047] In some embodiments, the second endothermic chamber **120** is configured to perform thermal oxygen-free decomposition on the feedstock to produce pyrolysis gases and vapors, and pyrolysis coke/ash. The second endothermic chamber **120** is configured to operate at a temperature of equal to or above 350° C., for example in the range of from 350° C. to 600° C. Thus, break-down of feedstock into various states of matter suitable for gasification in the first endothermic chamber **110** and for combustion in the exothermic chamber **130** can be performed in the second endothermic chamber **120**.

[0048] In some embodiments, a first channel **181** is connected between the external source **170** of feedstock to the inlet **172** of the exothermic chamber **130**. In some embodiments, the first channel **181** is further connected between the first outlet **114** of the first endothermic chamber **110** and the inlet **172** of the exothermic chamber **130**. In some embodiments, the first channel **181** is also connected between the upper outlet **124** of the second endothermic chamber **120** and the inlet **172** of the exothermic chamber **130**. Using the channel **181**, feedstock can be transferred from the external source **170**, the first endothermic chamber **110** and/or the second endothermic chamber **120** to the exothermic chamber **130** for combustion. In some embodiments, the first channel **181** is connected between the upper outlet **124** of the second endothermic chamber **120** and the first outlet **114** of the first endothermic chamber **110**. Using the channel **181**, feedstock can be transferred, for example, from the second endothermic chamber **120** to the first endothermic chamber **110**.

[0049] In some embodiments, the gasification device **100** further comprises a second channel **182** that is connected between the inlet **174** to the exothermic chamber **130** and the second outlet **115** of the first endothermic chamber **110**. The second channel **182** can further be connected between the inlet **174** to the exothermic chamber **130** and the lower outlet **125** of the second endothermic chamber **120**. Using the second channel **182**, feedstock can be transferred from the external source **170**, the first endothermic chamber **110** and/or the second endothermic chamber **120** to the exothermic chamber **130**. In some embodiments, the second channel **182** is connected between the lower outlet **125** of the second endothermic chamber **120** and the second outlet **115** of the first endothermic chamber **110**. Using the second channel **182**, feedstock can be transferred from the second endothermic chamber **120** to the first endothermic chamber **110** for gasification. Further using the second channel **182**, feedstock can also be transferred from the second endothermic chamber **120** to the combustion zone **131** of the exothermic chamber **130**.

[0050] In some embodiments, the inlet **172** is configured to receive feedstock in solid, liquid, decomposed and/or pyrolysis feedstock from the external source **170**. In some embodiments, the inlet **172** is configured to receive feedstock in solid, liquid state, decomposed feedstock and/or residue of gasification from the first outlet **114** of the first endothermic chamber **110** for combustion. In some embodiments, the inlet **172** is configured to receive feedstock in gaseous and/or vaporized state from the upper outlet **124** of the second endothermic chamber **120** for combustion. At least one effect can be is to feed the exothermic chamber **130** with the feedstock for combustion. Yet another effect can be is to remove the residual feedstock after gasification from the gasification device **100**.

[0051] In some embodiments, the first outlet **114** is configured to receive the feedstock in solid, liquid, decomposed and/or pyrolysis feedstock from the external source **170** for gasification. In some embodiments, the first outlet **114** is configured to receive pyrolysis feedstock in gaseous and/or vaporized state from the upper outlet **124** of the second endothermic chamber **120** for gasification. Thus, additional feedstock from the external source **170** can be fed into the first endothermic chamber **110** to produce a high yield of gas, preferably a high yield of synthesis gas (CO+H2). Further, upon pyrolysis in the second endothermic chamber **120**, processed or decomposed feedstock can be provided from the second endothermic chamber **120** to the first endothermic chamber **110**, in particular, to perform gasification of the processed or decomposed feedstock.

[0052] In some embodiments, the inlet **174** is configured to receive feedstock in gaseous or vaporized state from the second outlet **115** of the first endothermic chamber **110** for combustion. In some embodiments, the inlet **174** is configured to receive feedstock in solid, liquid, decomposed or pyrolysis feedstock from the lower outlet **125** of the second endothermic chamber **120**. In some embodiments, the second outlet **115** is configured to receive the feedstock in solid, liquid, decomposed and/or pyrolysis form from the lower outlet **125** of the second endothermic chamber **120** for gasification. The second endothermic chamber **120** being configured with the lower outlet **125** enables removal of pyrolysis ash from the second endothermic chamber **120**. Further, the first endothermic chamber **110** can be fed from the second endothermic chamber **120** with feedstock in solid and/or fluid phase for gasification.

[0053] In some embodiments, the gasification device **100** comprises an exhaust outlet **152** which is provided, for example, in an upper wall portion of the gasification device **100**. The exhaust outlet **152** is adapted to release waste heat from the gasification device **100**.

[0054] In the gasification device according to the example illustrated in FIG. **1**, the plurality of heat-transfer rods **140** can comprise one or more of a solid heat pipe, a hollow heat pipe and/or a convection heat pipe. In some embodiments, the plurality of heat-transfer rods are provided each as a solid heat pipe. In some embodiments, the plurality of heat-transfer rods are provided each as a hollow heat pipe. In some embodiments, the plurality of heat-transfer rods are provided each as a convection heat pipe.

[0055] Now, an exemplary heat-transfer rod comprised in the plurality of heat-transfer rods **140** will be described. The exemplary heat-transfer rod **140** comprises metal, for example, steel and/or a steel alloy. In some embodiments, the heat-transfer rod **140** comprises a core and a shell that, at least in a portion of the heat-transfer rod that is located in the combustion zone **131** of the exothermic chamber **130**, encloses the core. The core is comprised of core material that differs from shell material. The shell is comprised of shell material that has a higher melting point then the core material. For example, the shell material can be steel, in particular, stainless-steel, while the core material can be copper, a copper alloy, aluminum, or an aluminum alloy. The exemplary heat-transfer rod **140** is configured to withstand a temperature of more than 1100° C. More particularly, the exemplary heat-transfer rod **140** is designed to withstand a temperature in the range of from 800° C. to 1000° C. In some embodiments, the heat-transfer rod **140** comprises material that has a melting point sufficiently above 1100° C. so that the heat-transfer rod **140** keeps its structure rather than bending or otherwise deforming or disintegrating when exposed to a temperature that is prevalent in the combustion zone **131** during operation of the exothermic chamber **130**.

[0056] In some embodiments, the exemplary heat-transfer rod comprises a hollow heat pipe. A cavity inside the hollow heat pipe can be filled with medium suitable to perform convective heat transport. In some embodiments, the hollow heat pipe is filled with a working fluid. In some embodiments the cavity inside the hollow heat pipe is provided with an internal structure adapted to optimize convective heat flow and delivery of heat at predetermined portions of the heat-transfer rod to the wall of the hollow heat pipe.

[0057] The exemplary heat-transfer rod is adapted to withstand a temperature of more than 1100° C. In some embodiments, the exemplary heat-transfer rod is adapted to withstand a temperature in the range from 800° C. to 1000° C., while transferring heat from the combustion zone **131** of the exothermic chamber **130** to the first endothermic chamber **110** and the second endothermic chamber **120**.

[0058] In some embodiments, the exemplary heat-transfer rod **140** comprises a solid heat pipe. The solid heat pipe is made of a metallic or an alloy material. The solid heat pipe can have a smaller diameter than a hollow heat pipe. A hollow heat pipe, to achieve a same heat transfer, may require a larger diameter for filling the rod with fluid and for having the fluid flow inside the hollow heat pipe. Thus, the solid heat pipe reduces the need of extra resources to achieve the same or even better efficiency for heat transfer.

[0059] In some embodiments, as in the example illustrated in FIG. **1**, the plurality of heat-transfer rods **140**, arranged inside the exothermic chamber **130**, extend through the first endothermic chamber **110** and into the second endothermic chamber **120**. In some embodiments (not shown), the plurality of heat-transfer rods comprises a first set of at least one heat-transfer rod that extends into the first endothermic chamber **110**. Further, the plurality of heat-transfer rods comprises a second set of at least one heat-transfer rod that extends into the second endothermic chamber **120**. Thus, the plurality of heat-transfer rods **140** thermally couples the combustion zone **131** of the exothermic chamber **130** to the first endothermic chamber **110** and/or to the second endothermic chamber **120**.

[0060] The plurality of heat-transfer rods **140** extend from the combustion zone of the exothermic chamber **130**. The plurality of heat-transfer rods **140**, in a portion provided in the combustion zone **131**, are configured to withstand a temperature of at least 1100° C. The plurality of heat-transfer rods **140** are configured to transport heat from the combustion zone **131** to the first endothermic chamber **110**, wherein the first endothermic chamber **110** is adapted for operation at a temperature in the range of from 700° C. to 1100° C. The heat-transfer rods **140** are configured to further transport heat to the second endothermic chamber **120**, wherein the second endothermic chamber **120** is adapted to operate at a temperature in the range of from 350° C. to 600° C. Thus, the plurality of heat-transfer rods **140** can transfer process heat generated in the combustion zone **131** of the exothermic chamber **130** to the first endothermic chamber **110** and to the second endothermic chamber **120**. In particular, heat can efficiently be transferred from the combustion zone **131** to the inside of the first endothermic chamber **110** and on from the inside of the first endothermic chamber **110** to the inside of the second endothermic chamber **120**. In some embodiments, the heat-transfer rods **140** are provided with multiple segments that are stacked on one another to form the respective heat-transfer rod.

[0061] Accordingly, the gasification device **100** is configured to transfer the process heat not only by radiation and by convection from the exothermic chamber **130** to the first endothermic chamber **110** and/or to the second endothermic chamber **120**, but also by conduction of heat in the heat-transfer rods **140**. Convection provides for process heat, generated from the exothermic chamber **130** and rising by convection in the gasification device **100**, to heat an outside of the first endothermic chamber **110** and/or an outside of the second endothermic chamber **120**. In contrast, conduction of heat in the plurality of heat-transfer rods **140** provides for the inside of the first endothermic chamber **110** and/or the inside of the second endothermic chamber **120** to be heated. Thus, the rate of heat transfer inside the gasification device **100** is increased, whereby the gasification device **100** can be more efficiently perform gasification than a conventional device.

[0062] In operation of an embodiment of the gasification device as illustrated in FIG. **1**, combustible material is fed to the combustion zone **131** of the exothermic chamber **130**. The combustible material can comprise feedstock in the form of pyrolysis vapors, gases and/or pyrolysis coke or ash provided from the first endothermic chamber **110** and/or from the second endothermic chamber **120**. The combustible material is combusted in the combustion zone **131** of the exothermic chamber **130** and generates process heat, for example, at a temperature of above 1100° C.

[0063] From the combustion zone **131**, process heat rises up to the first endothermic chamber **110** which, for example, operates at a temperature in a range of from 700° C. to 1100° C. In the first endothermic chamber **110**, gasification of feedstock takes place to produce a hydrogen rich synthesis gas. In some embodiments, the gasification of feedstock produces a synthesis gas $CH4+H2O=CO+3\ H2$. From the first endothermic chamber **110**, process heat rises up further to the second endothermic chamber **120** which, for example, operates at a temperature in a range of from 350° C. to 700° C. In the second endothermic chamber **120**, feedstock is thermally decomposed without oxygen to produce pyrolysis gases and vapors, and pyrolysis coke/ash. The feedstock is broken down into various states of matter suitable for gasification in the first endothermic chamber **110** and/or for combustion in the exothermic chamber **130**.

[0064] In some embodiments, process heat rises up convectively from the exothermic chamber **130** to the outside of the first endothermic chamber **110** and to the outside of the second endothermic chamber **120**. Thus, process heat is provided to the first endothermic chamber **110** and to the second endothermic chamber **120**.

[0065] Further, the plurality of heat-transport rods **140** conduct process heat from the combustion zone **131** into the first endothermic chamber **110** and into the second endothermic chamber **120**. More particularly, the plurality of heat-transfer rods **140**, extending from the combustion zone **131** of the exothermic chamber **130** to the second endothermic chamber **120**, and having a lower temperature at the second endothermic chamber **120** than in the combustion zone **131** of the exothermic chamber **130**, conduct heat generated in the combustion zone **131**, following a temperature gradient along the plurality of heat-transfer rods **140**, to the second endothermic chamber **120**. In the first endothermic chamber **110**, the reformer process uses process heat from the combustion zone **131** to generate synthetic gas. In the second endothermic chamber **120**, the pyrolysis uses process heat from the combustion zone **131** to break-down feedstock for use in the reformer process performed in the first endothermic chamber **110**.

[0066] In some embodiments, the gasification device **100** further comprises a steam generator (not shown in FIG. **1**) and the at least one heat-transfer rod **140** is further adapted to transfer the process heat from the exothermic chamber **130** to the steam generator (not shown in FIG. **1**). The steam generator is configured to operate at a temperature of greater than 120° C. or a temperature range from 120° C. to 350° C. The at least one heat-transfer rod **140** extending from the exothermic chamber **130** with a temperature of at least 1100° C. cools down to the temperature in the range of from 120° C. to 350° C., before contacting the steam generator. The process heat is dissipated to the first endothermic chamber **110** operating at the temperature in the range of from 700° C. to 1100° C., and to the second endothermic chamber **120** operating at the temperature in the range of from 350° C. to 700° C. At least one effect can be that the residual process heat of the at least one heat-transfer rod **140** and/or heat rising in the exothermic chamber **130** is further utilized for generating steam internally in the device.

[0067] FIG. **2** is a drawing that schematically illustrates a sectional view of a gasification device **200** for gasification of feedstock according to some embodiments that implements a steam generator. The illustrated embodiment is analogous to the embodiments defined above with reference to FIG. **1**. Accordingly, the gasification device **200** comprises an exothermic chamber **230** that is provided with a combustion zone **231**. Further, the gasification device **200** comprises a first endothermic chamber **210**, a second endothermic chamber **220** and a plurality of heat-transfer rods **240**. In addition, the gasification device **200** comprises a steam generator **260**. Using the plurality of heat-transfer rods **240**, process heat generated in the combustion zone **231** of the exothermic chamber **230** can thus be conducted to the first endothermic chamber **210**, to the second endothermic chamber **220** and to the steam generator **260**. At least one effect of the steam generator **260**, when operated to generate steam, can be a reduction of steam that needs to be provided from a source external to the gasification device **200**.

[0068] In some embodiments, at least one of the first endothermic chamber **210**, the second endothermic chamber **220** and the steam generator **260** is arranged inside the exothermic chamber **230** of the gasification device **200**. For example, as illustrated in FIG. **2**, the second endothermic chamber **220** and the first endothermic chamber **210** are arranged inside the exothermic chamber **230**, above the combustion zone **231**, in a stack above one another, while the steam generator **260** is provided above the stack. In the exemplary embodiment, a ceiling wall **239** of the exothermic chamber **230** forms a floor wall of the steam generator **260**, whereby the steam generator **260** is thermally integrated with the gasification device **200**. Thus, the second endothermic chamber **220**, the first endothermic chamber **210** and the steam generator **260** can use process heat generated inthe exothermic chamber **230**. Thus, the first endothermic chamber **210**, the second endothermic chamber **220** and the steam generator **260** can carry out the endothermic reactions and steam generation, respectively, using process heat produced in the exothermic chamber **230**.

[0069] The steam generator **260** is coupled, via a steam outlet **269** connected by a steam channel **283** to a reformer inlet **216**, to the first endothermic chamber **210**. At least one effect can be that the steam which is produced within the gasification device **200** can be used in the first endothermic chamber **210** to produce a synthesis gas ($CO+H_2$), i.e., $CH_4+H_2O \rightleftarrows CO+3\,H_2$.

[0070] In an exemplary embodiment, the steam generator **260** comprises a water preheater **262**, a steam producer **263** and a steam super-heater **264**. The water preheater **262** is configured to use process heat to heat the water to a temperature in the range of from 120° C. to 500° C. In some embodiments, the water preheater **262** is configured to use process heat to heat the water to a temperature in the range of from 180° C. to 400° C. In some embodiments, the water preheater **262** is configured to use process heat to heat the water to a temperature in the range of from 240° C. to 300° C. The heated water can then be transferred to the steam producer **263**. In some embodiments, the water is kept as water above atmospheric pressure. In some embodiments, the water is kept at atmospheric pressure, whereby the process heat is used to convert the water to steam.

[0071] In some embodiments, the water preheater **262** of the steam generator **260** is coupled, via a water inlet **265**, to an external water source **279** for intake of water. In some embodiments, process heat can be used by the water preheater **262** to preheat the water taken in for use in the generation of steam. In particular, in an embodiment, the steam generator **260** is configured to release process heat from the gasification device **200**, via a first exhaust outlet **267**, to the external water source **279**. Thus, heat of exhaust gases can preheat water of the external water source **279**.

[0072] In some embodiments, the steam super-heater **264** is configured to produce steam using heat of synthesis gas produced by the first endothermic chamber **110**.

[0073] In some embodiments, a gas outlet **212** is configured to supply synthesis gas to the outside of the gasification device **200**. In some embodiments, the gas outlet **212** is coupled to the steam generator **260** to supply the synthesis gas from the first endothermic chamber **210** to the steam generator **260** for producing steam. In some embodiments, the gas outlet **212** is coupled to the super-heater **264** and configured to supply gas from the first endothermic chamber **210** to the super-heater **264** for producing steam. In some embodiments, the gas outlet **212** is coupled to the steam generator **260** to supply some part of the gas to the steam generator **260** for producing steam and some part to a gas container (not shown) for other applications of the synthesis gas, such as for running an engine. In some embodiments, the gas outlet **212** is coupled to a gas container (not shown) for supply of the synthesis gas outside the gasification device **200** to the container (not shown). At least one effect can be that the gas outlet **212** couples the first endothermic chamber **210** to the steam generator **260** and/or to the gas container (not shown) so that the gas usage is maximized and no gas is leaked or wasted to the surroundings.

[0074] The heat-transport rods **240** extend from the bottom of the exothermic chamber **230** and are adapted to dissipate process heat to the inside of the first endothermic chamber **210**, to the inside of the second endothermic chamber **220** and to the inside of the steam generator **260**. At least one effect can be that the plurality of heat-transfer rods **240** can conduct heat from the combustion zone **231** of the exothermic chamber **230** to the inside of the first endothermic chamber **210**, to the inside of the second endothermic chamber **220** and/or to the inside of the steam generator **260**.

[0075] In some embodiments, the plurality of heat-transfer rods **240** extends into the steam producer **263** for generating steam while using process heat dissipated from the plurality of heat-transfer rods **240**. The steam producer **263** is configured to convert the heated water to steam using process heat conducted by the plurality of heat-transfer rods **240**.

[0076] The inlet **222** is configured to receive feedstock. In some embodiments, the feedstock is supplied from any storage outside of the gasification device **200**. The second endothermic chamber **220** performs pyrolysis on the feedstock. In particular, the second endothermic chamber **220** performs thermal oxygen-free decomposition on the feedstock to produce pyrolysis gases and vapors, and pyrolysis coke/ash. The second endothermic chamber **220** operates at a temperature of greater than 350° C. or in the range of from 350° C. to 600° C. At least one effect of the second endothermic chamber **220** can be to.

[0077] In some embodiments, the second endothermic chamber **220** may be modified according to the operational parameters of the gasification device **200**, like temperature, pressure etc.

[0078] The steam generator **260** comprises a water preheater **262**, a steam producer **263** and/or a steam super-heater **264**. At least one effect can be that the exhaust gas from the gasification device **200** is used to run the steam generator **260**, when the steam generator **260** is integrated with the gasification device **200**.

[0079] Accordingly, the gasification device **200** is configured to transfer the process heat from the exothermic chamber **230** to the first endothermic chamber **210**, to the second endothermic chamber **220** and/or to the steam generator **260** by two channels. The first channel is to heat the outside of the first endothermic chamber **210**, the second endothermic chamber **220** and/or the steam generator **260** by the process heat, generated from the exothermic chamber **230**, rising in the gasification device **200**. The second channel is to heat the inside of the first endothermic chamber **210**, the second endothermic chamber **220** and/or the steam generator **260** by the heat-transfer rods extending from the exothermic chamber **230**. At least one effect of both the channels of heat transfer can be to increase the rate of heat transfer inside the device, whereby making the device more efficient for the process of gasification.

[0080] In operation of an embodiment of the gasification device as illustrated in FIG. **2**, combustible material is fed to the combustion zone **231** of the exothermic chamber **231**. The combustible material can comprise feedstock in the form of pyrolysis vapors, gases and/or pyrolysis coke or ash provided from the first endothermic chamber **210** and/or from the second endothermic chamber **220**. The combustible material is combusted in the combustion zone **231** of the exothermic chamber **230** and generates process heat, for example, at a temperature of above 1100° C.

[0081] From the combustion zone **231**, process heat rises up to the first endothermic chamber **210** which, for example, operates at a temperature in a range of from 700° C. to 1100° C. In the first endothermic chamber **210**, gasification of feedstock takes place to produce a hydrogen rich synthesis gas. In some embodiments, the gasification of feedstock produces a synthesis gas $CH_4+H_2O \rightleftarrows CO+3\ H_2$. One example of the produced synthesis gas is a gas comprising H_2 and CO components. In another example, the produced synthesis gas comprises CO_2, O_2, CH_4 and/or N_2 components. From the first endothermic chamber **210**, process heat rises up further to the second endothermic chamber **220** which, for example, operates at a temperature in a range of from 350° C. to 700° C. In the second endothermic chamber **220**, feedstock is thermally decomposed without oxygen to produce pyrolysis gases and vapors, and pyrolysis coke/ash. The feedstock is broken down into various states of matter suitable for gasification in the first endothermic chamber **210** and for combustion in the exothermic chamber **230**.

[0082] From the second endothermic chamber **220**, process heat rises up still further to the steam generator **260** which, for example, operates at a temperature in a range of from 120° C. to 350° C.

[0083] In some embodiments, the process heat rises up convectively from the exothermic chamber **230** to the outside of the first endothermic chamber **210**, to the outside of the second endothermic chamber **220** and to the steam generator **260**. Thus, the process heat warms up the first endothermic chamber **210**, the second endothermic chamber **220** and the steam generator **260**.

[0084] Further, the plurality of heat-transport rods **240** conduct process heat from the combustion zone **231** into the first endothermic chamber **210**, into the second endothermic chamber **220** and into the steam generator **260**. More particularly, the plurality of heat-transfer rods **240**, extending from the combustion zone **231** of the exothermic chamber **230** to the steam generator **260**, and being colder at the steam generator **260** than in the combustion zone **231**, conduct heat generated in the combustion zone **231** of the exothermic chamber **230**, following a temperature gradient along the plurality of heat-transfer rods **240**, to the steam generator **260**. The steam generator uses the process heat from the exothermic chamber **230** in the generation of steam. The steam is directed from the steam generator **260** into the first endothermic chamber **210** where the steam is used to perform the reformer process to produce synthesis gas.

[0085] Although specific embodiments have been illustrated and described herein, it will be appreciated by those of ordinary skill in the art that a variety of alternate and/or equivalent implementations may be substituted for the specific embodiments shown and described without departing from the scope of the present invention. This application is intended to cover any adaptations or variations of the specific embodiments discussed herein. For example, while in the examples illustrated in FIG. **1** and in FIG. **2** the gasification device comprises a plurality of heat-transfer rods, in another example (not shown) the gasification device merely comprises a single heat-transfer rod.

[0086] Although some drawings may be provided with exemplary dimensional statements, it should be understood that such statements are merely exemplary. Neither can any such statements be understood to be consistent from one drawing to another, nor should the statements be understood to limit the scope of the present disclosure to the stated dimensions or any combination or ratio thereof. The disclosure of dimensions should be understood merely to state an order of magnitude according to some embodiments. In particular, the invention can be implemented using, other orders of magnitude, other dimensions, other ratios of dimensions. Further, it should be understood that drawings are not drawn to scale.

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[0087] The implementations herein are described in terms of exemplary embodiments. However, it should be appreciated that individual aspects of the implementations may be separately claimed and one or more of the features of the various embodiments may be combined.

Claims

1. A gasification device, the gasification device comprising: an exothermic chamber provided with a combustion zone that is configured to perform combustion to produce process heat, a first endothermic chamber adapted to perform gasification of feedstock; and a second endothermic chamber adapted to subject feedstock to an endothermic break-down reaction; wherein the gasification device is configured such that the first endothermic chamber uses combustion process heat to perform the gasification process, and wherein the gasification device is configured such that the second endothermic chamber uses combustion process heat to subject feedstock to the endothermic break-down reaction; the gasification device further comprising at least one heat-transfer rod adapted to transfer combustion process heat from the combustion zone of the exothermic chamber to the first endothermic chamber and to the second endothermic chamber.

2. The gasification device of claim 1, wherein the at least one heat-transfer rod is arranged inside the exothermic chamber and extends into the first endothermic chamber and into the second endothermic chamber.

3. The gasification device of claim 1, wherein the at least one heat-transfer rod comprises a solid heat pipe, a hollow heat pipe or a convection heat pipe.

4. The gasification device for gasification of feedstock of claim 1, wherein at least two of the second endothermic chamber, the first endothermic chamber and the combustion zone of the exothermic chamber are arranged above one another.

5. The gasification device of claim 1, wherein at least one of the second endothermic chamber and the first endothermic chamber is arranged inside the exothermic chamber.

6. The gasification device of claim 1, wherein the first endothermic chamber is a reformer.

7. The gasification device of claim 6, wherein the reformer is configured to comprise a fluidized reformer-bedchamber.

8. The gasification device of claim 1, wherein the first endothermic chamber includes a reformer inlet configured to be coupled to a source of steam.

9. The gasification device of claim 8, the device further comprising a steam generator configured to receive water and to generate steam from the water, wherein the steam generator comprises a steam outlet connected to the reformer inlet.

10. The gasification device of claim 9, wherein the at least one heat-transfer rod is further adapted to transfer combustion process heat from the combustion zone of the exothermic chamber to the steam generator.

11. The gasification device of claim 1, wherein the second endothermic chamber is a pyrolysis reactor.

12. The gasification device of claim 11, wherein the pyrolysis reactor comprises a fluidized reactor-bedchamber.

13. The gasification device of claim 1, wherein the combustion zone of the exothermic chamber is configured to combust fuel to produce combustion process heat.

14. The gasification device of claim 1, wherein the exothermic chamber comprises an inlet configured to receive feedstock, and wherein the combustion zone of the exothermic chamber is configured to combust the feedstock.

15. The gasification device of claim 1, wherein the exothermic chamber further comprises a bottom inlet configured to be coupled to an oxygen source, and/or wherein the exothermic chamber further comprises a bottom outlet configured to release ash from the exothermic chamber.